

13 - Intro to organic chemistry

23/12/24

Organic chemistry is the study of organic compounds

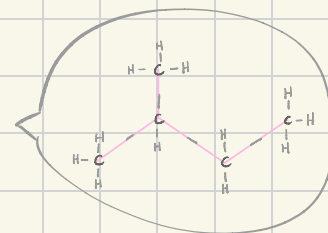
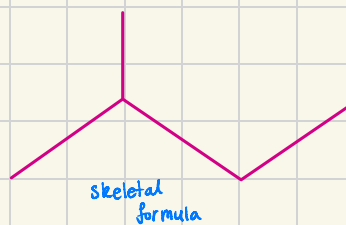
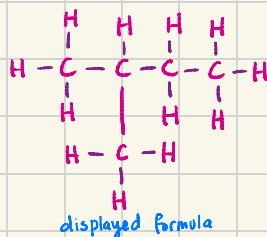
→ compounds that contain carbon (& hydrogen)

Formulae for organic compounds

- i) structural formula → shows the order of the atoms joined together in a compound
- ii) displayed formula → shows the order of the atoms joined together & their orientation/bond angles
- iii) skeletal formula → simplifies carbon & hydrogen & only shows functional groups

e.g. 2-methylpentane

structural formula
 $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$



Functional groups

* a functional group is a group of atoms in an organic compound that determine the chemical properties of the compound

→ a homologous series is a series of compounds with the same functional group & they differ in the number of carbon atoms

Homologous series	Functional group	prefix/suffix	structural formula
alkene	C=C bond	-ene	$\begin{matrix} R & & R \\ & \diagdown & / \\ & C = C \\ & / & \diagdown \\ R & & R \end{matrix}$ rest of the compound
halogenoalkane	halogen	halo- depends on the halogen	$\text{R}-\text{X}$ any halogen
alcohol	hydroxyl	-ol	$\text{R}-\text{OH}$
aldehyde	carbonyl	-al	$\begin{matrix} & & \text{O} \\ & & \\ \text{R}- & \text{C} & \\ & & \text{H} \end{matrix}$
ketone	carbonyl	-one	$\begin{matrix} & \text{O} \\ & \\ \text{R}- & \text{C} & -\text{R}' \end{matrix}$
carboxylic acid	carboxyl	-oic acid	$\begin{matrix} & \text{O} \\ & \\ \text{R}- & \text{C} & \\ & & \text{OH} \end{matrix}$
ester	ester	-oate	$\text{R}-\text{C}-\text{O}-\text{C}(=\text{O})-\text{R}$

eg.	name	displayed formula	skeletal formula
	propene	$\begin{matrix} & \text{H} & \text{H} \\ & & \\ \text{H}- & \text{C} & -\text{C} = \text{C} \\ & & \\ & \text{H} & \text{H} \end{matrix}$	
	if X → chlorine 1-chloropropane	$\begin{matrix} & \text{H} & \text{H} & \text{X} \\ & & & \\ \text{H}- & \text{C} & -\text{C} & -\text{C}-\text{H} \\ & & & \\ & \text{H} & \text{H} & \text{H} \end{matrix}$	
	propan-1-ol	$\begin{matrix} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}- & \text{C} & -\text{C} & -\text{C}-\text{O}-\text{H} \\ & & & \\ & \text{H} & \text{H} & \text{H} \end{matrix}$	
	propanal	$\begin{matrix} & \text{H} & \text{H} & & \text{O} \\ & & & & \\ \text{H}- & \text{C} & -\text{C} & -\text{C} & \\ & & & & \text{H} \end{matrix}$	
	propanone	$\begin{matrix} & \text{H} & \text{O} & \text{H} \\ & & & \\ \text{H}- & \text{C} & -\text{C} & -\text{C}-\text{H} \\ & & & \\ & \text{H} & & \text{H} \end{matrix}$	
	propanoic acid	$\begin{matrix} & \text{H} & \text{H} & & \text{O} \\ & & & & \\ \text{H}- & \text{C} & -\text{C} & -\text{C} & \\ & & & & \text{OH} \end{matrix}$	
	methyl propanoate	$\begin{matrix} & \text{H} & \text{O} & \text{H} & \text{H} \\ & & & & \\ \text{H}- & \text{C} & -\text{O}-\text{C} & -\text{C} & -\text{C}-\text{H} \\ & & & & \\ & \text{H} & & \text{H} & \text{H} \end{matrix}$	

amine	amine	-amine	$R-NH_2$
nitrile	nitrile	-nitrile	$R-C \equiv N$

propylamine	$\begin{array}{c} H & H & H \\ & & \\ H-C & -C & -C-N \\ & & \\ H & H & H \end{array}$	
propanenitrile	$\begin{array}{c} H & H \\ & \\ H-C & -C & -C \equiv N \\ & \\ H & H \end{array}$	

24/12/24

Hydrocarbons

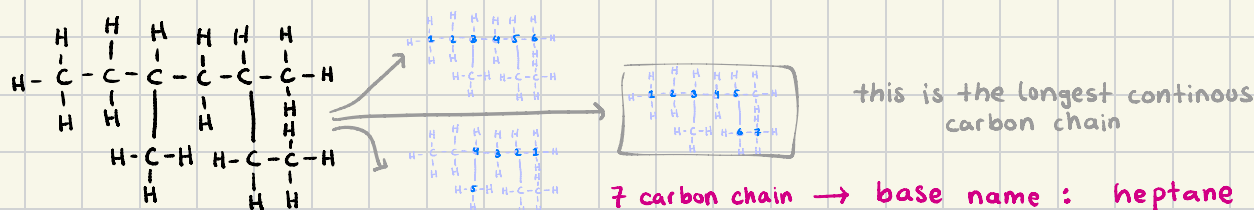
→ compounds made up of carbon & hydrogen atoms

→ alkanes are hydrocarbons with no functional group

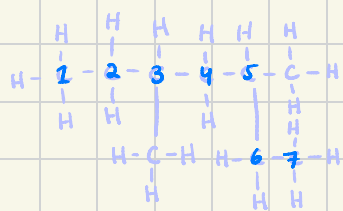
Naming compounds

• compounds are named using systematic nomenclature (IUPAC naming)

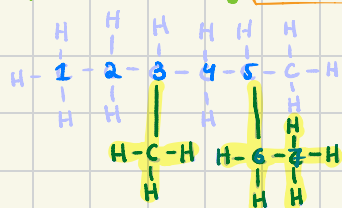
Step 1: identify the longest carbon chain (parent chain) - determines the base name of compound



Step 2: number the parent chain



Step 3: identify **substituents** ← groups of atoms that branch off of the main carbon chain



at carbon #3 there is a methyl group ($-CH_3$)

at carbon #5 there is an ethyl group ($-CH_2CH_3$)

Step 3: name the compound

parent chain: heptane

substituents: methyl at carbon #3 & ethyl at carbon #5] substituents are arranged alphabetically
→ ethyl and then methyl

Name: - **5-ethyl-3-methylheptane**

Saturated compounds

compounds that only have single bonds

Unsaturated compounds

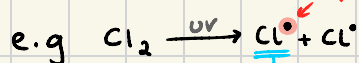
compounds that contain double/triple bonds

Fission

- a process through which chemical bonds are broken, resulting in the formation of two or more species
- two kinds: homolytic and heterolytic

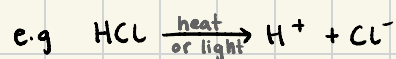
covalent bond breaks evenly & each atom involved in the bond retains one of the two electrons that were previously shared in the bond

represents a free radical



free radical: species with an unpaired electron

covalent bond breaks unevenly & one of the atoms takes both of the electrons that were originally shared in the bond. This results in a formation of a cation & anion



★ steps of free radical reaction:

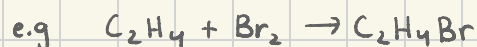
- 1) initiation: formation of the free radicals - energy is required to break the covalent bonds
- 2) propagation: radicals attack the reactant molecules to generate more free radicals
- 3) termination: two radicals meet & form a product molecule, leading to the end of reaction

free radical substitution: reaction where a free radical replaces an atom/group in a molecule
* often seen in the halogenation alkanes

Types of organic reactions

i) addition reactions

formation of a single product from two or more reactants



ii) elimination reactions

removal of a small molecule from a larger reactant molecule

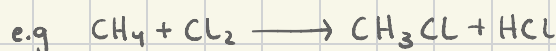


iii) condensation reactions

reactions where addition & then elimination take place e.g. $\text{CH}_3\text{OH} + \text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$

iv) substitution reactions

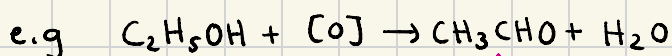
involve the replacement of an atom(s) by another



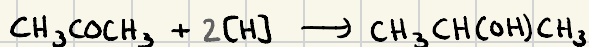
Oxidation: loss of electrons or increase in oxidation state, involving the addition of oxygen or removal of hydrogen

Reduction: gain of electrons or decrease in oxidation state, involving the removal of oxygen or addition of hydrogen

in organic redox reactions, [O] represents an atom of oxygen from an oxidising agent & [H] represents an atom of hydrogen from a reducing agent



ethanol loses two hydrogens & has thus been oxidised



* don't forget to balance equation

Nucleophiles

25/12/24

- an electron-rich species that donates electrons

nucleophilic reactions: reactions where nucleophiles donate electrons

nucleophilic substitution: reaction where a nucleophile replaces a leaving group in a molecule

nucleophilic addition: reaction where a nucleophile adds to a molecule with polar bonds

both result in the formation of a new covalent bond

Electrophiles

- an electron-deficient species that accepts electrons

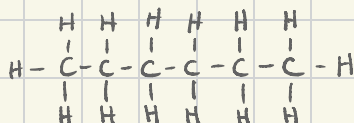
electrophilic reactions: reactions where electrophiles accept electrons

electrophilic addition: reaction where an electrophile adds to a molecule with double/triple bonds

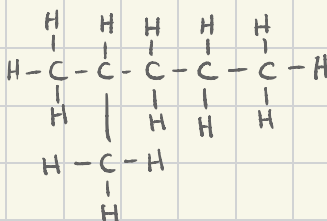
Shapes of organic molecules

- can either be

straight-chained

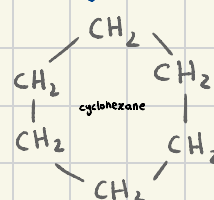


branched



or

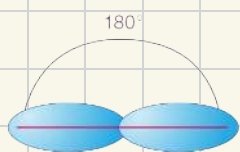
cyclic



- in hybridised atoms

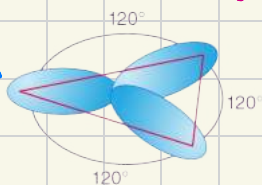
sp

linear shape with a 180° bond angle



sp^2

trigonal planar shape with a 120° bond angle



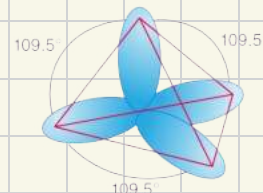
sp & sp^2 atoms are all planar

made of:

- 1 sigma (σ) bond
- 2 pi (π) bonds

sp^3

tetrahedral shape with a bond angle of 109.5°



made of:

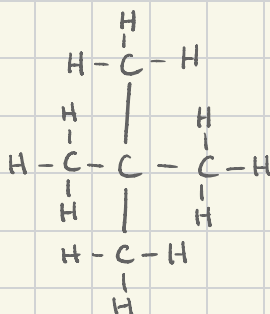
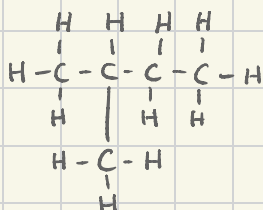
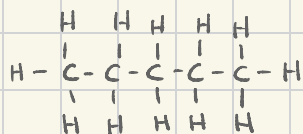
only sigma (σ) bonds

26/12/24

- ## Structural isomerism

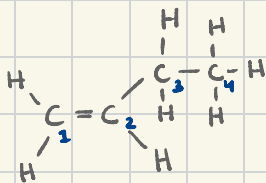
three types

- ↳ different arrangements of the carbon skeleton

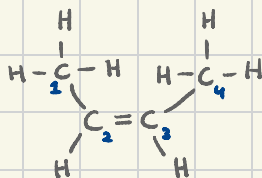


all 3 have the
molecular formula: C_5H_{12}

- variation in position of functional group / substituent



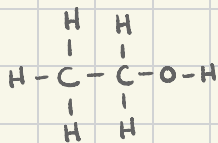
double bond in 1-2



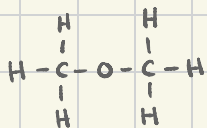
double bond in 2-3

both have the
molecular formula: C_4H_8

- ↳ different functional groups within compound



alcohol



ether
(don't need to know)

both have the
molecular formula: C_2H_6O

Stereoisomerism

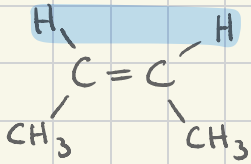
* compounds with the same molecular formula but a different arrangement of their spatial orientation

two types

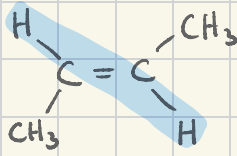
1) geometrical isomerism

↳ occurs in molecules with restricted rotation (due to a double bond; presence of a π bond)

↳ two kinds: cis (substituents on the same side) $\neq$ trans (substituents on opposite sides)



cis-2-buten

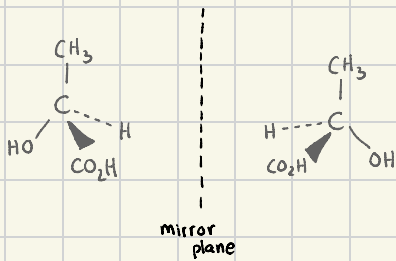


trans-2-buten

2) optical isomerism

↳ occurs when two molecules are mirror images of each other but cannot be superimposed

↳ exist because of chiral centres in a molecule



these are optical isomers,
referred to as enantiomers.

a chiral centre refers to a carbon atom bonded
to four different groups

chiral centres
give rise to enantiomers

* compounds can contain
more than one chiral centre